



BORDER WATCH: INTELLIGENT MULTI-STATION SURVEILLANCE AND OBJECT CORRELATION SYSTEM

CSA0201 – C PROGRAMMING

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INTRODUCTION



- C-based simulation of a multi-station border surveillance system.
- Multiple stations send object detections to a central command center.
- Detects whether incoming reports represent new or already detected objects.
- Uses TCP, pthreads, dynamic memory, and linked lists for system operation.
- Generates logs and alerts for important surveillance events.



PROBLEM STATEMENT



- Multiple stations may detect the same object, creating duplicate records.
- Independent detections can lead to inconsistent object information.
- Concurrent station requests may cause race conditions in shared data.
- Important detections require reliable logging and alert communication.
- A unified system is needed to correlate and manage multi-station detections.



LITERATURE SURVEY



s.no	Authors & Year	Title	Key Contribution	Relevance to Project
1	Kumar et al., 2024	AI-Based Border Surveillance System	Used AI and computer vision for real-time border monitoring and intrusion detection.	Supports AI-based monitoring and detection in Border Watch.
2	Singh et al., 2024	Smart Border Monitoring Using IoT and AI	Integrated IoT sensors with CCTV for continuous surveillance.	Supports combining sensors and cameras for border monitoring.
3	Patel et al., 2025	AI-Driven Border Surveillance and Threat Detection	Used YOLO and real-time video analytics to detect potential threats.	Relevant to real-time object and threat detection.
4	Raj et al., 2025	Automated Border Intrusion Detection System	Combined GPS, IoT, and image analysis for intrusion detection.	Supports location-based intrusion alerts in the project.
5	Ahmed et al., 2025	Smart Surveillance for Secure Border Management	Used edge computing to reduce processing delay and improve response time.	Relevant for fast, real-time alerts in Border Watch.



OBJECTIVES



- Develop a multi-station client-server surveillance system.
- Identify and merge duplicate object detections.
- Maintain a dynamically growing object database.
- Ensure safe concurrent access using mutex synchronization.
- Generate logs and communicate critical alerts through IPC.



MODULE 1 – MULTI-STATION DETECTION & OBJECT CORRELATION



- Surveillance stations communicate with the command center using TCP sockets.
- Each detection contains location, timestamp, characteristics, and confidence.
- Incoming detections are compared with existing objects.
- Spatial and temporal thresholds identify duplicate detections.
- Matching detections are merged into a single master object record.



MODULE 2 – SYNCHRONIZED COMMAND CENTER & RESPONSE



- Command center manages objects using dynamic memory and linked lists.
- Pthreads and mutexes enable safe concurrent processing.
- Recursive traversal manages the Sector → Subsector → Post zone hierarchy.
- Events are recorded in an append-only log file.
- Critical objects generate alerts through System V message queues.



CORE TECHNOLOGIES



1

TCP Sockets

Station-to-command-center
communication

2

Pthreads & Mutexes

Safe concurrent request
processing

3

Dynamic Memory & Linked Lists

Growing object database

4

System V IPC

Critical alert message queues



ADVANTAGES & LIMITATIONS



Advantages

- Reduces duplicate object records.
- Supports concurrent multi-station processing.
- Provides synchronized and dynamic object management.
- Maintains an audit trail through logging.

Limitations

- Uses simulated detection data instead of real sensors.
- Basic thresholds may not handle every real-world condition.
- Linked-list searching may become slower with very large datasets.



FUTURE ENHANCEMENTS



- Integrate real cameras, radar, GPS, or IoT sensors.
- Implement advanced multi-object tracking algorithms.
- Add a graphical dashboard for live surveillance monitoring.
- Introduce encrypted station-to-server communication.
- Use scalable databases and intelligent anomaly detection.



CONCLUSION



- BorderWatch demonstrates a complete multi-station surveillance simulation in C.
- It combines TCP networking, multithreading, dynamic memory, and IPC.
- Spatial-temporal correlation converts multiple detections into consistent object records.
- Synchronization and logging improve reliability and traceability.
- The system provides a strong foundation for developing a scalable real-time surveillance platform.

THANK YOU



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Thursday, 27/08/2026 10:49 AM GMT +05:30